Hub: PHY180 Hub
Previous: 7 — 1st and 2nd Laws of Newton
Next: 9 — Friction and Centripetal Force

Referenced Textbook Sections

5.5-5.7

Lecture Notes

See related In-Lecture Addendums

Pre-Lecture Notes

Newton's Third Law

Newton's Third Law generally is defined as the Law of Reaction, that being that every force has an equal and opposite reactive force applied on them across bodies:

> Whenever a body exerts a force on a second body, the first body experiences a force equal in magnitude and opposite in direction to the force it exerts.

For this context, and analyzing systems using NTL—it may help to separate the bodies of the system to try and properly understand where the reactive forces are being applied.

In-Lecture Addendum

Newton's Third Law always operates in related force-reaction force pairs. These force-reaction force pairs also *act on opposite bodies*.

In-Lecture Addendum 2

Note that this extends to non-contact forces (such as Newtonian Gravitation).

Common Forces

This section will detail the commonly seen forces present in classical mechanics.

> Normal Force
> The force experienced on an object lying on a surface, as a force to resist the object's weight from causing it to fall through said surface. May be less than the weight of the actual object itself (e.g. on an incline),

> Friction
> The force of friction is generally defined as a relationship between a coefficient of friction, and an object's weight , that is $F_F = \mu F_N$.
> Friction will be more properly defined in later notes.

> Spring Force
> The force applied by a spring (or more generally, the restorative force by any material in the linear-elastic regimen) can be generally applied by Hooke's Law:
>
> $$F_E = -k\Delta x$$
>
> Note the negative sign, this force is a restorative force opposite the direction of displacement.